

# Step 7 Decision-Matrix Results

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## 0.1 Step 7: Decision-Matrix Results

This note summarizes the implemented Step 7 results from the pooled-reference susceptibility workflow. The current run uses the two available representation families:

- **name\_only**
- **concat**, which is the Step 7 label used for the Step 6 **name\_description** representation

The family **fusion** is not available in the current pipeline outputs. The pooled-reference usability parameters were fixed *ex ante* at  $\lambda_{\min} = 0.90$  and  $\iota_{\max} = 0.10$ .

**Main pooled-reference result** For both available representation families, the admissible-maximizer set was empty under the current usability rule. Consequently, the selected pooled cutoff in each case is the least aggressive susceptibility maximizer rather than the largest admissible maximizer.

Table 1: Pooled-reference Step 7 decision matrix for the all-tracks graph

| Criterion                | Direction     | <b>name_only</b>           | <b>concat</b>              | Better family |
|--------------------------|---------------|----------------------------|----------------------------|---------------|
| Selected cutoff $\tau^*$ | n/a           | 0.585136                   | 0.519089                   | n/a           |
| Selection rule           | n/a           | least aggressive maximizer | least aggressive maximizer | tie           |
| Retained pooled edges    | higher better | 379                        | 739                        | <b>concat</b> |
| Connected components     | lower better  | 438                        | 259                        | <b>concat</b> |
| Largest-component share  | higher better | 0.0791                     | 0.2681                     | <b>concat</b> |
| Isolate share            | lower better  | 0.4812                     | 0.2534                     | <b>concat</b> |
| Connected-node share     | higher better | 0.5188                     | 0.7466                     | <b>concat</b> |
| Max susceptibility       | higher better | 8.6303                     | 19.6044                    | <b>concat</b> |

The pooled all-tracks graph is fragmented for both families under the current rule, but **concat** retains substantially more usable structure than **name\_only**. At the pooled level, **concat** dominates **name\_only** on all substantive topological criteria reported above.

Table 2: Comparison-safe pooled-reference summaries by unit type

| Family           | Unit type             | $n$ graphs | Mean density | Mean LCC share | Mean isolate share | Mean connected share |
|------------------|-----------------------|------------|--------------|----------------|--------------------|----------------------|
| <b>name_only</b> | <b>all_tracks</b>     | 1          | 0.001364     | 0.079088       | 0.481233           | 0.518767             |
| <b>name_only</b> | section               | 13         | 0.008771     | 0.126077       | 0.683614           | 0.316386             |
| <b>name_only</b> | track                 | 2          | 0.002140     | 0.061127       | 0.551507           | 0.448493             |
| <b>name_only</b> | programme             | 14         | 0.014638     | 0.157464       | 0.637584           | 0.362416             |
| <b>name_only</b> | <b>programme_year</b> | 33         | 0.016891     | 0.172514       | 0.703674           | 0.296326             |
| <b>concat</b>    | <b>all_tracks</b>     | 1          | 0.002659     | 0.268097       | 0.253351           | 0.746649             |
| <b>concat</b>    | section               | 13         | 0.016973     | 0.230276       | 0.472074           | 0.527926             |
| <b>concat</b>    | track                 | 2          | 0.004159     | 0.196186       | 0.330775           | 0.669225             |
| <b>concat</b>    | programme             | 14         | 0.031873     | 0.274761       | 0.375935           | 0.624065             |
| <b>concat</b>    | <b>programme_year</b> | 33         | 0.050945     | 0.295730       | 0.443752           | 0.556248             |

Across all exported comparison-safe unit types, the pooled-reference sparse graphs are denser and less isolated under **concat** than under **name\_only**. The contrast is especially strong for tracks, programmes, and programme-year units.

Table 3: Local-diagnostic summaries by unit type

| Family                 | Unit type      | $n$ graphs | Mean local density | Mean local LCC share | Mean local isolate share | Mean pooled-ref density in unit |
|------------------------|----------------|------------|--------------------|----------------------|--------------------------|---------------------------------|
| <code>name_only</code> | section        | 13         | 0.055955           | 0.503787             | 0.138484                 | 0.008771                        |
| <code>name_only</code> | track          | 2          | 0.005527           | 0.238120             | 0.319429                 | 0.002140                        |
| <code>name_only</code> | programme      | 14         | 0.097655           | 0.518424             | 0.190085                 | 0.014638                        |
| <code>name_only</code> | programme_year | 33         | 0.129193           | 0.506373             | 0.182349                 | 0.016891                        |
| <code>concat</code>    | section        | 13         | 0.060244           | 0.453927             | 0.136751                 | 0.016973                        |
| <code>concat</code>    | track          | 2          | 0.006774           | 0.306356             | 0.207894                 | 0.004159                        |
| <code>concat</code>    | programme      | 14         | 0.079115           | 0.408992             | 0.212149                 | 0.031873                        |
| <code>concat</code>    | programme_year | 33         | 0.129426           | 0.529329             | 0.169554                 | 0.050945                        |

**Interpretation** The local-diagnostic graphs are materially less fragmented than the pooled-reference induced graphs, which is expected because each local graph receives its own threshold. However, these local graphs are retained only for robustness purposes and are not the comparison-safe basis for downstream analyses.

In substantive terms, the current Step 7 results point in the same direction across the pooled graph and the lower-level summaries: `concat` preserves more structure, isolates fewer nodes, and yields higher connected-node shares than `name_only`. This is not yet the formal Step 8 representation decision, but it is the main empirical signal carried forward from Step 7.